Marxian Competitive Dynamics Simulator

Technical Documentation, Assumptions, and User Guide

Version 2.0

Based on Marx, Capital Vol. I (Ch. 10, 15) and Vol. III (Ch. 13–15)

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Contents

1. Purpose and Overview 4
2. Theoretical Framework 4
2.1 The Labor Theory of Value 4
2.2 Output as a Function of All Production Inputs 4
2.3 Price-Taking Firms and Market Allocation 5
2.4 The Organic Composition of Capital 5
2.5 The Tendency of the Rate of Profit to Fall 6
2.6 The Working Day and Physiological Limits 6
3. Model Structure and Formulas 7
3.1 Variable Capital (v) 7
3.2 Output 7
3.3 Capital Stock (C) and Annual Depreciation 7
3.4 Cost Per Widget 7
3.5 Market Allocation and Units Sold 8
3.6 Financials 8
4. Assumptions and Modeling Decisions 9
4.1 Core Assumptions 9
4.2 Key Modeling Decisions 9
Output grounded in all four production inputs 9
Price-taking firms, not price-setting 10
Unsold widgets as a real cost 10
Deriving C from v rather than setting it independently 10
No wage rate 10
Rate of profit on capital stock, not revenue 10
Excel export 10
5. User Inputs: Reference Guide 12
5.1 Market Inputs 12
Market price / widget 12
Market size 12
5.2 Firm-Level Inputs 12
Workers 13
Hours per day 13
Days per week 14

Tech multiplier 14
Capital lifespan 14
6. Output Metrics Reference 16
6.1 Labor 16
6.2 Capital Composition 16
6.3 Production & Market 16
6.4 Per Widget 17
6.5 Annual 17
7. Interpretive Notes and Scenarios 18
7.1 The Core Competitive Dynamic 18
7.2 Volume vs. Cost Competition 18
7.3 The Overproduction Dynamic 18
7.4 Extending the Working Day as a Substitute for Technology 18
7.5 The Rate of Profit Paradox 18
7.6 Capital Lifespan and the Rate of Profit 19

1. Purpose and Overview

This tool models the competitive dynamics of capitalist production as described by Karl Marx in Capital, Volumes I and III. It is designed to make the abstract mechanisms of Marxian political economy tangible and interactive: specifically, the process by which technology investment displaces living labor, drives down the cost of producing a unit of output, and generates competitive market advantage — while simultaneously raising the organic composition of capital and suppressing the rate of profit across the industry.

The simulator is not a full macroeconomic model. It operates at the level of two competing firms within a single commodity market, holding demand constant (inelastic). Its purpose is pedagogical and analytical: to demonstrate the structural logic of competitive accumulation and its consequences for workers, firms, and profitability.

The tool draws on three interlocking bodies of analysis from Capital:

- Volume I, Chapter 10: The Working Day — the struggle over the length of the working day as a site of class conflict and the physiological limits of labor extraction.
- Volume I, Chapter 15: Machinery and Large-Scale Industry — how mechanization increases relative surplus value, displaces workers, and compels the extension of the working day among remaining workers.
- Volume III, Chapters 13–15: The Law of the Tendency of the Rate of Profit to Fall — how the rising organic composition of capital (C/v) suppresses the rate of profit even as the mass of surplus value grows.

2. Theoretical Framework

2.1 The Labor Theory of Value

The model is grounded in Marx's labor theory of value. Value is created only by living labor — human beings working. Capital (machinery, technology, fixed assets) does not create new value; it transfers its own value to the product as it is consumed in production. This is the distinction Marx draws between variable capital (v) and constant capital (C).

Variable capital is "variable" because it is the source of surplus value: workers produce more value than they are paid. Constant capital is "constant" because it merely transfers pre-existing value without generating new value. This distinction is the foundation of Marx's critique of profit and the rate of profit.

2.2 Output as a Function of All Production Inputs

A central design principle of this simulator is that output is grounded in actual production inputs. Every widget produced is a direct product of how many workers are employed, how long they work each day, how many days per week they work, and the productivity multiplier conferred by technology:

```
output = workers × hours/day × days/week × 52 × tech multiplier
```

Each labor-hour produces exactly one widget at a tech multiplier of 1. The tech multiplier scales output per labor-hour proportionally. This grounds the model in the materialist logic of production: you cannot produce without labor and capital working together, and the quantity produced is determined by the quantity and intensity of both inputs.

This is a deliberate departure from models that treat output as an assumption or that assign market share automatically based on efficiency. In this model, market share is earned through production volume.

2.3 Price-Taking Firms and Market Allocation

Both firms are price takers. The market price is set exogenously and applies equally to both firms. Neither firm has pricing power. This is consistent with Marx's treatment of competitive commodity markets, where the socially necessary labor time for production establishes the market value of the commodity, and individual firms must produce at or below that cost to survive.

Market share is determined proportionally by each firm's share of total combined output:

```
output share = firm output / (Firm A output + Firm B output) units sold
= min(firm output, market size × output share)
```

If total combined output is less than or equal to market size, each firm sells everything it produces. If combined output exceeds market size, each firm sells its proportional share of market demand — and incurs the cost of unsold widgets with no revenue return. A firm whose cost per widget exceeds the market price cannot cover costs and exits production entirely, producing zero.

2.4 The Organic Composition of Capital

The organic composition of capital is the ratio of constant capital to variable capital: C/v. As firms invest in labor-saving technology, C rises relative to v. In this model:

```
C (total capital stock) = tech multiplier × v × capital lifespan
Organic composition (C/v) = tech multiplier × capital lifespan
```

A firm with a tech multiplier of 7 and a 10-year capital lifespan has an organic composition of 70 — it has advanced 70 units of capital stock for every unit of variable capital. This is the structural feature Marx argues drives the tendency of the rate of profit to fall.

2.5 The Tendency of the Rate of Profit to Fall

The rate of profit uses the Vol. III stock-based formulation — total capital advanced, not annual revenue:

```
rate of profit = s / (C + v) where s = surplus value = revenue - total
annual cost
```

As C rises relative to v — driven by firms adopting labor-saving technology to gain competitive advantage — the denominator grows faster than the numerator. The individual firm gains a short-term competitive advantage by reducing cost per unit. But as the technology generalizes across the industry, all firms see their organic composition rise, and the industry-wide rate of profit falls. Individual rationality produces collective self-undermining of profitability.

Critically: profit is calculated on total annual cost (including the cost of any unsold widgets), not just the cost of units sold. A firm that produces more than the market absorbs incurs a real loss on unsold inventory even if its per-widget margin is positive.

2.6 The Working Day and Physiological Limits

Marx dedicates an entire chapter of Volume I to the working day because it is the terrain on which absolute surplus value is extracted — surplus produced by extending the duration of labor beyond what is necessary to reproduce the worker's own subsistence. The tool expresses labor in physiological terms: hours per day and days per week, rather than aggregate annual hours. This makes the human cost of labor intensity immediately legible.

Extending hours or days is a substitute for technology investment as a means of increasing output volume. However, it does not reduce cost per widget, because cost and output scale together. The physiological thresholds are flagged explicitly: amber above 10 hrs/day or 50 weekly hours; red above 12 hrs/day or 60 weekly hours.

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3. Model Structure and Formulas

3.1 Variable Capital (v)

```
v = workers × hours/day × days/week × 52
```

v is the total annual living labor index. It is not expressed in dollars — this is a deliberate modeling choice. Introducing a wage rate would require assumptions about nominal vs. real wages, white-collar vs. blue-collar compensation, benefits, and regional variation, all of which are secondary to the structural logic being demonstrated. The index approach keeps the focus on the quantity of labor time, which is what Marx's value theory is fundamentally about.

3.2 Output

```
output = v × tech multiplier = workers × hours/day × days/week × 52 ×
tech multiplier
```

Each labor-hour produces one widget at multiplier = 1. The tech multiplier scales that proportionally. All four production inputs — workers, hours per day, days per week, and tech multiplier — multiply together. No single input dominates; output is the product of all four. This is the corrected formulation from earlier versions of the model, which treated output as independent of production inputs.

3.3 Capital Stock (C) and Annual Depreciation

```
C = tech multiplier × v × capital lifespan annual depreciation = C /
lifespan = tech multiplier × v annual production cost = depreciation +
v = (tech multiplier + 1) × v
```

Capital stock is derived from the variable capital base, the tech multiplier, and the capital lifespan. This keeps the model parsimonious and preserves the internal logic that rising tech investment always implies a rising organic composition. Annual depreciation is the annual flow of capital value transferred to the product. The full capital stock C enters the denominator of the rate of profit calculation, representing total capital advanced.

3.4 Cost Per Widget

```
cost/unit = annual production cost / output          = ((tech
multiplier + 1) × v) / (v × tech multiplier)          = (tech
multiplier + 1) / tech multiplier          = 1 + (1 / tech multiplier)
```

This algebraic simplification reveals that cost per widget is purely a function of the tech multiplier — independent of the number of workers, hours per day, or days per week. Adding workers or extending hours does not change the unit cost of production. Only the tech multiplier determines competitive position on cost. More workers produce more widgets but at exactly the same cost per widget.

This is a key theoretical result: it means that labor intensity (more workers, longer hours) is a strategy for volume and revenue, not for competitive cost advantage. Only capital investment can lower the cost floor.

3.5 Market Allocation and Units Sold

```
total supply = Firm A output + Firm B output  (profitable firms only)
firm share = firm output / total supply units sold = min(firm output,
market size × firm share)
```

If total supply ≤ market size: each firm sells its entire output. If total supply > market size: each firm sells in proportion to its share of total supply, and unsold widgets represent a real cost with no revenue. A firm with cost/unit > market price exits and produces zero — it cannot participate in production at all.

3.6 Financials

```
revenue = market price × units sold total annual cost = annual
production cost  (incurred on full output, not just units sold) unsold
widget cost = cost/unit × unsold widgets surplus value (s) = revenue −
total annual cost profit = s profit margin = profit / revenue rate of
profit = s / (C + v)     [Vol. III stock-based formulation]
```

Profit is calculated against total annual cost, not against the cost of units sold only. This means a firm producing beyond what the market absorbs is penalized for its overproduction — the cost of unsold widgets is a real drain on profit. This is the mechanism by which excess productive capacity suppresses profitability even for a firm with a per-widget margin advantage.

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4. Assumptions and Modeling Decisions

4.1 Core Assumptions

- Inelastic demand. Market size is fixed regardless of price. This eliminates demand-side effects and isolates competitive cost dynamics.
- No wage rate. Labor is expressed as a quantity index (hours), not dollars. This focuses the model on the structural ratio of living labor to capital rather than nominal compensation dynamics.
- Single commodity market. Both firms produce identical, undifferentiated widgets. No product differentiation, branding, or quality competition.
- Proportional market allocation. Market share is determined by each firm's share of total combined output — not assumed from cost advantage alone. Both firms are price takers.
- Firms produce to capacity. There is no demand-side constraint on production decisions. Firms produce what their inputs allow. If total supply exceeds market size, unsold output is a loss.
- No financial structure. The model excludes interest, debt service, taxes, retained earnings, and dividends. Profit equals surplus value.
- No input cost variation. Raw material prices, energy costs, and supply chain factors are not modeled. All cost variation flows through labor and capital.
- Straight-line depreciation. Capital stock depreciates evenly over its lifespan. No accelerated depreciation, write-downs, or residual value.
- No learning curves. The tech multiplier is fixed and does not improve over time with use or scale.
- Two-firm duopoly. The model does not simulate oligopoly dynamics, entry, exit equilibrium, or industry-wide convergence beyond the two firms shown.
- Market exit is instantaneous. A firm whose cost/unit exceeds the market price produces nothing and earns no revenue. There is no gradual wind-down or below-cost survival on reserves.

4.2 Key Modeling Decisions

Output grounded in all four production inputs

Earlier versions of this model treated output as independent of production inputs, with market share assigned automatically based on cost advantage. This was corrected. Output is now the direct product of workers × hours/day × days/week × 52 × tech multiplier. Market share falls out of production volume, not from assumed competitive dominance. This is the only formulation consistent with the model's materialist premises.

Price-taking firms, not price-setting

The market price is exogenous. Neither firm controls it. Both firms respond to it. This is consistent with Marx's treatment of competitive commodity markets and eliminates the temptation to frame one firm as strategically dominant. The structural advantage of lower unit cost expresses itself through survival, not through pricing strategy.

Unsold widgets as a real cost

When total supply exceeds market demand, firms do not simply produce less — they incur the cost of unsold output with no revenue return. This is a deliberate choice to make the overproduction dynamic visible even within a model that does not formally analyze crisis theory. The unsold widget cost line and utilization rate give a clear signal when productive capacity has outrun market absorption.

Deriving C from v rather than setting it independently

Capital stock is derived as C = mult × v × lifespan rather than entered as a free parameter. This preserves the internal logic that technology investment is always proportional to the labor base it is replacing, and keeps the parameter space manageable for a pedagogical tool.

No wage rate

The decision to express v as labor hours (not wages) is theoretically motivated. Marx's value theory operates in units of socially necessary labor time, not money. Introducing wages would require taking a position on the relationship between the wage and the value of labor power — a relationship Marx treats as itself a site of exploitation and struggle. The hour index is more theoretically honest for this model's purposes.

Rate of profit on capital stock, not revenue

Using s/(C+v) rather than profit/revenue as the rate of profit measure is a deliberate departure from conventional accounting. The accounting margin measures how efficiently a firm converts sales into profit. The Marxian rate of profit measures the return on total capital advanced — which drives investment decisions at the macroeconomic level and which Marx argues is subject to a structural tendency to fall as capital deepens.

Excel export

The tool includes a one-click Excel export via SheetJS, running entirely in the browser with no server required. The export captures all inputs and all output metrics for both firms in a single sheet, organized in the same section order as the on-screen cards. Percentage metrics are formatted as percentages; large integers receive comma separators; unit cost figures display four decimal places for precision. The export timestamp is embedded so each snapshot is self-documenting.

5. User Inputs: Reference Guide

Each firm (A and B) has five independent inputs. The market has two global inputs. All inputs have both a slider and a direct number entry field that stay synchronized.

5.1 Market Inputs

Input	Range / Default	Description & Impact
Market price / widget	0.5–20, no default	The price at which every widget sells. Exogenous to both firms — neither firm sets it. Expressed in cost index units. The revenue ceiling for both firms.
Market size (widgets/yr)	1k–10M, default 500k	Total annual demand for widgets, fixed and inelastic. Determines how many units the market will absorb from the combined output of both firms.

Market price / widget

- Increasing: raises revenue and surplus value for profitable firms; widens the margin between revenue and cost; raises the rate of profit; keeps more firms profitable; reduces competitive pressure to reduce costs.
- Decreasing: compresses margins; simulates the competitive price destruction Marx describes. When market price crosses a firm's cost/unit threshold from above, that firm exits. When it crosses both firms' thresholds, both exit — crisis of valorization.
- Key dynamic: dragging this slider down is the primary way to simulate the long-run competitive pressure that eliminates less-capitalized producers from the market.

Market size

- Increasing: amplifies absolute revenue and profit for all profitable firms without affecting cost/unit or competitive position. Makes the stakes of production volume higher.
- Decreasing: reduces the revenue base. If market size shrinks below total combined output, utilization rates fall and unsold widget costs rise for both firms.
- Interaction with output: when combined firm output exceeds market size, a supply glut emerges. Both firms sell less than they produce and incur unsold costs. This is the embryo of the overproduction dynamic.

5.2 Firm-Level Inputs

Input	Range / Default	Description & Impact
Workers	1–500 / A:100, B:40	Number of workers employed. Sets the scale of v and all derived quantities. Does not affect cost per widget.
Hours per day	1–24 / A:8, B:10	Length of the working day. Multiplies into total labor hours. Physiological warnings activate above 10 hrs/day (amber) and 12 hrs/day (red).
Days per week	1–7 / A:5, B:6	Working days per week. Multiplies into annual labor hours. Combined with hours/day determines weekly labor burden per worker.
Tech multiplier	1–20 / A:2, B:7	Productivity per labor-hour. A multiplier of 7 means each labor-hour produces 7 widgets. Determines cost/unit and organic composition. The primary competitive variable.
Capital lifespan (yrs)	1–20 / A:5, B:10	Years the capital stock remains in service. Longer lifespan inflates C relative to annual flows, suppressing the rate of profit without affecting cost/unit or competitive position.

Workers

- Increasing: raises total output proportionally. More output → larger share of total supply → more units sold and more revenue. Does not change cost per widget. Does not change competitive position on cost.
- Decreasing: reduces output and revenue scale proportionally. Firm produces fewer widgets and earns smaller revenue, but at the same unit cost and margin.
- For workers: more workers means more people exposed to the working conditions set by the other inputs. It also means more total living labor (v), which raises C (capital stock) proportionally through the derivation formula.
- Important: workers and tech multiplier interact. A firm with 200 workers and a multiplier of 2 produces the same output as a firm with 100 workers and a multiplier of 4 — but the latter has half the labor force, twice the organic composition, and a lower cost per widget.

Hours per day

- Increasing: raises total annual labor hours, raises v, raises output, raises revenue — all proportionally. Does not change cost per widget. The firm produces more but at no cost advantage.
- Decreasing: reduces all flow quantities proportionally.
- For workers: the most direct expression of the working day in Marx's sense. Above 10 hrs: elevated injury risk, burnout, reduced cognitive function. Above 12 hrs: chronic health damage, shortened life expectancy. Above 16 hrs: physiological breakdown territory. The model flags these thresholds visibly but does not enforce them as

constraints — reflecting the historical reality that capital has repeatedly pushed past these limits when legally permitted.

- Key insight: extending hours extracts more absolute surplus value in aggregate (more output, more revenue) but does not lower the cost of production per unit. Only the tech multiplier can do that.

Days per week

- Increasing: same structural effect as increasing hours/day — raises total annual labor, output, and revenue proportionally. Does not change cost/unit or competitive position.
- Decreasing: reduces total annual labor hours and all derived quantities. A four-day week reduces v by 20% relative to a five-day week.
- For workers: combined with hours/day, determines the total weekly labor burden. Weekly hours = hours/day × days/week. Above 50 weekly hours: elevated strain. Above 60: severe overwork with documented chronic disease outcomes.
- Six and seven-day weeks appear in the model as strategies to increase output volume without improving unit economics — a labor intensification strategy that extracts more absolute surplus but does not generate competitive cost advantage.

Tech multiplier

- Increasing: lowers cost per widget (cost/unit = 1 + 1/multiplier), increases output per labor-hour, raises output volume, and raises C relative to v. The only input that simultaneously increases productivity and decreases unit cost.
- Decreasing: raises cost per widget. If cost/unit rises above market price, the firm exits.
- For workers: higher multiplier means each worker produces more per hour — but also means fewer workers are needed to produce the same output. This is the mechanism of technological unemployment: the reserve army of labor that Marx argues suppresses wages and weakens worker bargaining power.
- For competition: the tech multiplier is the decisive competitive variable. A firm that achieves a multiplier advantage gains a structural cost advantage that more workers and longer hours cannot overcome. This is the engine of capital accumulation: the race to invest in productivity-enhancing technology, which drives the organic composition upward across the industry.
- Rate of profit paradox: raising the tech multiplier raises absolute surplus value (more output, lower cost/unit) but also raises C dramatically (C = mult × v × lifespan). The rate of profit s/(C+v) may fall even as absolute profit rises. This is the Marxian paradox: the most technologically advanced firms tend toward the lowest rates of profit on total capital advanced.

Capital lifespan

- Increasing: raises total capital stock C without changing annual depreciation or cost per unit. Competitive position is unchanged. But the denominator of the rate of profit (C + v) grows substantially, suppressing s/(C+v). Longer-lived capital is a structural drag on the rate of return.
- Decreasing: reduces capital stock, raises the rate of profit. Short lifespan implies rapid obsolescence — machinery replaced frequently — more characteristic of software or light manufacturing than of heavy industry.
- For workers: no direct effect on working conditions in this model. Indirectly, a firm with large long-lived capital stock has made a major fixed commitment that it will be motivated to utilize intensively — pressure toward longer hours, more shifts, and minimum downtime.
- Theoretical significance: capital lifespan isolates the distinction between the annual flow of capital cost (depreciation) and the total stock of capital advanced. Two firms can have identical annual depreciation charges but very different capital stocks if their lifespans differ. The rate of profit on total investment — not the annual margin — is what determines whether the capital commitment was worthwhile. This is the Vol. III insight: high-capital, long-lifespan industries tend structurally toward lower rates of profit.

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6. Output Metrics Reference

6.1 Labor

Output metric	What it means
Weekly hrs / worker	Hours/day × days/week. Color-coded: amber above 50, red above 60.
Annual hrs / worker	Weekly hours × 52. Total annual labor time per worker.
Widgets / worker / day	= hours/day × tech multiplier. Individual daily productivity. Rises with tech multiplier; unaffected by number of workers or days per week.
Widgets / day (firm)	Widgets/worker/day × workers. Total daily firm output.
Widgets / week (firm)	Widgets/day × days/week. Total weekly firm output.

6.2 Capital Composition

Output metric	What it means
v — variable capital	Total annual living labor index (workers × annual hrs/worker).
C — total capital stock	mult × v × lifespan. Total capital advanced.
Annual depreciation	C / lifespan = mult × v. Annual capital value transferred to product.
Annual cost (depr + v)	(mult + 1) × v. Total annual production cost incurred regardless of sales.
Organic composition (C/v)	= mult × lifespan. Ratio of total capital stock to annual variable capital. Key structural indicator of capital intensity.
Labor share of annual cost	v / (depreciation + v). Falls as tech multiplier rises.

6.3 Production & Market

Output metric	What it means
Annual output (widgets)	workers × annual hrs/worker × tech multiplier. What the firm actually produces. This is the grounded production figure — all four inputs determine it.
Output share of supply	Firm output / total combined output of profitable firms. Determines units sold allocation.
Units sold	min(firm output, market size × output share). What the market absorbs from this firm.
Utilization rate	Units sold / annual output. Below 100% when total supply exceeds market size. Color-coded: green ≥90%, amber ≥50%, red below 50%.
Unsold widgets	Annual output − units sold. Widgets produced but not sold. Their cost is fully incurred.

Output metric	What it means
Cost index / widget	1 + 1/tech multiplier. The firm's unit cost of production. If this exceeds market price, the firm exits.
Market price / widget	The exogenous market price. Same for both firms.
Market status	Active if cost/unit ≤ market price. Market exit if cost/unit > market price.

6.4 Per Widget

Output metric	What it means
Revenue / widget	Equal to market price.
Profit / widget (sold)	Total annual profit / units sold. Note: this spreads the full annual cost (including unsold widget cost) across sold units, so it may be negative even if market price exceeds cost/unit when utilization is low.
Margin / widget	(Market price − cost/unit) / market price. Pure per-unit margin, independent of volume.

6.5 Annual

Output metric	What it means
Revenue (index)	Market price × units sold.
Total annual cost (index)	Annual production cost incurred on full output — not just units sold.
Unsold widget cost	Cost/unit × unsold widgets. The penalty for overproduction.
Surplus value (s)	Revenue − total annual cost. Includes the drag of unsold inventory.
Profit (index)	Equal to surplus value in this model.
Profit margin	Profit / revenue. Green ≥15%, amber ≥0%, red if negative.
Rate of profit s/(C+v)	Surplus value / (total capital stock + annual variable capital). The Marxian rate. Suppressed by rising C. Tends to fall as capital deepens even when absolute profit grows.

7. Interpretive Notes and Scenarios

7.1 The Core Competitive Dynamic

Set Firm A with a low-tech multiplier (2–3) and Firm B with a high multiplier (7–10). Firm B will have a lower cost/unit. Firm B's larger output-per-labor-hour will give it a larger share of total supply. Now drag the market price slider down. Watch the point at which the market price crosses Firm A's cost/unit threshold: Firm A exits. Continue dragging until it crosses Firm B's threshold: both exit — crisis of valorization.

7.2 Volume vs. Cost Competition

Give both firms the same tech multiplier but different numbers of workers. The firm with more workers produces more widgets and captures a larger share of the market — but at the same cost per unit and the same margin. Revenue scales, but so does cost. Now change the tech multiplier for one firm: watch the cost/unit diverge immediately. This demonstrates the distinction between volume competition (workers, hours) and cost competition (tech multiplier). Only the latter generates structural advantage.

7.3 The Overproduction Dynamic

Set both firms with high worker counts and high-tech multipliers. Watch combined annual output exceed market size. Utilization rates fall, unsold widget costs appear, and profit margins compress even for the firm with the lower cost/unit. This is the embryo of the Marxian overproduction crisis: competitive accumulation drives productive capacity beyond what the market can absorb, and profitability suffers industry-wide even before any firm is technically unprofitable on a per-unit basis.

7.4 Extending the Working Day as a Substitute for Technology

Give both firms the same tech multiplier. Increase Firm A's hours per day to 12 or 14. Cost/unit does not change. Firm A produces more in aggregate and earns more revenue if the market absorbs it, but gains no cost advantage. The physiological flags activate. This demonstrates Marx's argument: extending the working day extracts more absolute surplus value but does not lower the cost of production per unit. Only technology can do that.

7.5 The Rate of Profit Paradox

Give Firm B a very high tech multiplier (15+) and a long capital lifespan (15–20 years). Observe that despite capturing a large share of the market and generating substantial absolute surplus value, the rate of profit is low — often single digits — because C is enormous relative to s. The firm is highly profitable in absolute terms but offers a poor return on the capital invested. This is

the paradox Marx describes: the most technologically advanced firms tend toward the lowest rates of profit.

7.6 Capital Lifespan and the Rate of Profit

Hold all inputs constant and vary only the capital lifespan for one firm. Watch the rate of profit fall as lifespan increases, even though cost/unit, output, market share, and absolute profit are unchanged. This isolates the suppressive mechanism: the denominator (C + v) grows while the numerator (s) does not. Industries with very long-lived capital — steel, mining, rail, utilities — tend structurally toward lower rates of profit than software or consumer goods. This model makes that structural feature directly observable.

Source texts: Marx, K. Capital: A Critique of Political Economy, Volumes I–III. Harvey, D. The Limits to Capital (1982). Shaikh, A. Capitalism: Competition, Conflict, Crises (2016).